Exact Diffusion Design for Maximally Collective Stable Turing Patterns

in Binary-Complex Mass-Action Networks

Topology-wide all-spectrum localization and exponent-optimal stationary heterogeneity trade-offs

Concise theorem summary — external audit draft

Family/realization space. For every $m \geq 3$, the binary-complex mass-action topology $\widehat{N}_m$ has species $X_1, \ldots, X_m, Z$ and reactions

$$0 \to X_1, \quad X_1 + X_i \to X_1 + X_{i+1} \ (2 \leq i \leq m-2),$$

$$X_1 + X_{m-1} \to 2X_m, \quad 2X_m \to X_2, \quad 2Z \rightleftarrows X_1 + X_m.$$

The chain range is empty for $m = 3$. With $n = m+1$, there are n species and $n+1$ reactions. The semipositive conservation vector is $c = (0, 4, \ldots, 4, 2, 1)^T$, $\operatorname{rank} \Gamma_m = m$, and

$$\ker \Gamma_m = \operatorname{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

The exact two-dimensional kernel and the $m+2$ columns give the stated rank by rank–nullity; the maximal minor on rows $X_1, X_3, \ldots, X_m, Z$ and columns $R_2, \ldots, R_{m-2}, R_a, R_b, R_+$ has determinant $4(-1)^m$. Every positive steady flux is $(a\mathbf{1}_m, b, b)$, and every positive-equilibrium Jacobian is exactly $J_m(a, b, H) = A_m(a, b)H$ with $a, b > 0$ and $H \succ 0$ diagonal. Conversely every such triple is physically realized.

Topology-wide localization. For every positive realization and every principal species set I with $0 < |I| < m$, $J_{I,I}$ is Hurwitz. Every nontrivial strongly connected diagonal block is one of two stable long cycles, a principal block of the boundary triad $\{X_1, X_m, Z\}$, or a negative singleton. The triad satisfies the cubic Routh–Hurwitz inequalities coefficientwise. For $m = 3$, the two cycles $\{X_1, X_2\}$ and $\{X_2, X_3\}$ and all remaining principal subgraphs are checked directly; the chain classification is used for $m \geq 4$. In contrast,

$$(-1)^m \det J_{\{X_1,\ldots,X_m\},\{X_1,\ldots,X_m\}} = -2a^{m-1}b \prod_{i=1}^{m} h_i < 0.$$

Hence, throughout every positive realization,

$$\min \{|I| : \varnothing \neq I \subseteq [n],\ \alpha(J_{I,I}) > 0\} = m = n - 1.$$

Every smaller subsystem is stable against both real and complex instability.

Principal-minor diffusion-ray theorem. Let $n \geq 2$ and $a_I = (-1)^{|I|} \det J_{I,I}$. Assume $\det J = 0$, $a_I > 0$ for $|I| \leq n-2$, and $\sum_{|I|=n-1} a_I > 0$ (in particular these coefficient hypotheses follow from a simple zero and otherwise stable spectrum together with stable lower principal blocks). Then for $D \succ 0$

$$\det(sD - J) = s(\beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}), \qquad \beta_k > 0 \ (k \geq 2).$$

A nonzero stationary threshold exists iff $\beta_1 < 0$; it is unique and simple. Moreover $J - sD$ has a positive real eigenvalue exactly for $0 < s < s_*$. This does not rule out nonreal wave instability outside that band. The network application is a one-bad-minor corollary, not a hypothesis of the general theorem. Here scalar simplicity is first proved for the dispersion root; ordinary algebraic simplicity of zero for $J - s_*D$ follows separately from $\chi'_{s_*}(0) > 0$.

Exact network diffusion law and sharp linear contrast. The complete order-$(n-1)$ omission table has one negative term (omit Z), positive weight 16 when an interior X_j is omitted, and zero when X_1 or X_m is omitted. Thus

$$\boxed{d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j}}$$

is necessary and sufficient for a stationary threshold on the ray sD. For every H in the explicitly defined homogeneously stable domain, the sharp nonattained contrast infimum is

$$T(H) = 8h_Z \sum_{j=2}^{m-1} h_j^{-1};$$

at $H = I$ it is $8(m-2)$. Every stationary crossing of this topology obeys $\chi_D \chi_H > 8(m-2)$. Here χ_H measures equilibrium concentration-scale separation; it is not interpreted as a universal measure of biological expense or implausibility.

Here $c^T J = 0$ and homogeneous stability on $c^\perp$ imply rank $n-1$, an algebraically simple conservation zero, and a positive order-$(n-1)$ characteristic coefficient. Thus the general diffusion-ray hypotheses hold. The criterion is independent of a, b, but the threshold is $s_*(a, b, H, D)$ and generally depends on them.

Exact unit-equilibrium stable design. At $a = b = 1$, $H = I$, write $K_i = 91m - 181 - i$ and take

$$d_1 = \frac{23}{63}, \quad d_i = K_i^{-1} \ (2 \le i \le m-1), \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}.$$

Write the full critical vectors as $r = (r_1, \ldots, r_m, r_Z)^T$ and $\ell = (\ell_1, \ldots, \ell_m, \ell_Z)^T$. The exact contrast is $23(91m - 183)/63$. A 35-term homogeneous half-plane certificate and a 77-term spatial certificate exclude every nonzero closed-right-half-plane first-mode root. Algebraic simplicity at the equality point follows from

$$\frac{d}{d\lambda} \det(\lambda I - A_m + D_m)\bigg|_{\lambda=0} = -\frac{163}{45}\ell^T r > 0.$$

Thus the first-mode kernel and cokernel are one-dimensional; exact critical vectors also give transversality $\eta_m > 0$. The homogeneous base case $m = 3$ is checked from $q_3 = (\lambda + 7)(\lambda^2 + 5\lambda + 2)$; the 35-term root exclusion applies for $m \ge 4$.

On the fixed integrated-mass class, the quadratic field satisfies

$$A_m w_0 = -\tfrac{1}{4}B(r,r), \quad c^T w_0 = 0, \qquad (A_m - 4D_m)w_2 = -\tfrac{1}{4}B(r,r).$$

The cubic coefficient is

$$c_m = \frac{\ell^T\{B(r,w_0) + \frac{1}{2}B(r,w_2)\}}{\ell^T r}.$$

This is the one-dimensional center-manifold normal-form coefficient in the adjoint-projection coordinate; setting the reduced vector field to zero gives the stationary Lyapunov–Schmidt equation. Spatial reflection $\xi \mapsto \pi - \xi$ sends $r\cos\xi$ to its negative and acts as $A \mapsto -A$ on an equivariant center manifold. Hence the reduced field is odd and the two nonzero branches are exchanged by reflection. The stationary fixed-mass linearization is Fredholm of index zero, with kernel $\operatorname{span}\{r\cos\xi\}$, cokernel $\operatorname{span}\{\ell\cos\xi\}$, and nonzero transversality pairing $(\pi/2)\ell^T D_m r$. A four-factor recurrence and a shifted-polynomial certificate prove $c_m < 0$ for every $m \ge 3$. The two supercritical positive branches are locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space.

Stable diffusion–equilibrium trade-off. Let $\nu = m - 2$,

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \ge 2, \end{cases} \qquad L_0 = \kappa_\nu/\sqrt{\nu}, \qquad L_1 = 90\nu/(90\nu + 1),$$

and for $L \in [L_0, L_1]$ set

$$h_1 = h_m = h_Z = 1, \qquad h_i = \frac{K_i}{LK_{i-1}}, \qquad \mathsf{H}_m(L) = \operatorname{diag}(h), \qquad D_m^{\text{phys}}(L) = \mathsf{H}_m(L)\Delta_m,$$

where Δ_m is the effective unit profile above. The physical equilibrium is $\mathsf{H}_m(L)^{-1}\mathbf{1}$. Under $\widehat{x} = \mathsf{H}_m(L)x$, the scaled PDE retains the parameter as $\partial_t \widehat{x} = \mathsf{H}_m(L)\{f_m(\widehat{x}) + (1-\mu)\Delta_m \partial_{\xi\xi}\widehat{x}\}$; at $\mu = 0$ its stationary bracket and critical right vector are unchanged. The interval is nonempty in every dimension. The transformed left vector $\widetilde{\ell}(L) = \mathsf{H}_m(L)^{-1}\ell$ satisfies

$$\widetilde{\ell}(L)^T \mathsf{H}_m(L)\Delta_m r = \ell^T \Delta_m r < 0, \qquad \widetilde{\ell}(L)^T r < 0,$$

so the crossing coefficient is positive. The kernel of the row-scaled first-mode operator remains $\operatorname{span}\{r\}$; the nonzero pairing $\widetilde{\ell}(L)^T r$ excludes a generalized eigenvector, so the scaled zero is algebraically simple. A 22-term homogeneous certificate for $\nu \ge 2$, a direct Routh–Hurwitz calculation for $\nu = 1$, an 84-term spatial certificate, and a gauge-corrected cubic comparison prove a primary supercritical branch that is locally exponentially asymptotically stable in the fixed-integrated-mass H^1 phase space for every certified L. The physical contrasts are

$$\chi_D(L) = \frac{23}{63}\,91\nu L, \qquad \chi_H(L) = \frac{91\nu - 1}{91\nu L}, \qquad \chi_D\chi_H = \frac{23(91\nu - 1)}{63}.$$

The product is independent of L and equals the unit-equilibrium stable-profile contrast,

$$\chi_D(L)\chi_H(L) = \chi_D^{\text{unit}}(m) = \frac{23(91m - 183)}{63}.$$

Thus L reallocates a fixed contrast product and reduces $\max(\chi_D, \chi_H)$, not the product. At $L = L_0$,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \qquad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu}\frac{91\nu - 1}{91\nu},$$

so both are $\Theta(\sqrt{m})$. The lower endpoint is the boundary of the present sufficient homogeneous modulus estimate for $\nu \ge 2$; it is not a proved intrinsic dynamical boundary. Since every stationary crossing of $\widehat{N}_m$ satisfies $\max(\chi_D, \chi_H) > \sqrt{8(m-2)}$, exponent $1/2$ is optimal among stationary-crossing realizations of this topology. No constant-optimal or complete Pareto-frontier claim is made.

PDE conclusion. On the real fixed-integrated-mass H^1 space, positive Neumann diffusion is sectorial and the quadratic mass-action Nemytskii map is smooth into L^2. The conservation restriction removes the homogeneous zero and the Neumann interval has no continuous translation mode. Strict branch-spectrum negativity therefore yields local exponential convergence for sufficiently close nonnegative initial data in the same mass class.

Most failure-prone points. External checking should focus on: (i) exhaustion of principal SCCs, including $b = 2a$; (ii) strict equality cases in the 22/35/77/84-term certificates and the direct $\nu = 1$ cubic; (iii) the principal-minor derivative monotonicity; (iv) the equilibrium-scaled fixed-mass gauge; and (v) the cubic comparison $N_m(L) > 1/200$.

Scope. The family is synthetic, rank deficient with one semipositive conservation law, and all-mobile. Results are local for each fixed dimension. No arbitrary-data global boundedness, global attraction, constant-optimal stable frontier, biochemical realization, cross-diffusion, or immobile-species theorem is claimed.